



< Previous



Next >

Complexity of objects within sets

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Problem: In a collection of objects, do the elements have the same complexity?

On many occasions, the complexity of objects is easy to estimate, just because these objects are already stored in the computer's memory. Consider, for instance, the alphabet: a, b, c,..., z. It can be retrieved as a list. In Python as in most computer languages, letters can be addressed through their ASCII code:

```
>>> Alphabet = [chr(c) for c in range(97,123)]
>>> print(Alphabet)
['a','b','c','d','e','f','g','h','i','j','k','l','m','n','o','p',
'q','r','s','t','u','v','w','x','y','z']
```

Once the alphabet is made available as a list, each letter can be retrieved by its rank on the list. So its complexity is estimated by the length of the binary representation of its rank.

For instance, one can designate the letter k by mentioning its position: 11 within the alphabet. An estimate of its complexity is 4, as we can code 11 with 4 bits (1011).

Question: Isn't it easier just to say 'k' instead of 1011?

Answer: No. 'k' is just a convenient representation of an ASCII code, 107, which is 1101011 in binary form.

Here is what we are able to do so far:

When an object can be retrieved from a list,
its complexity within the list can be estimated
by the length of the binary representation of its
rank.

Ordered list

1/1 point (ungraded)

About 57 000 people took part in the Paris marathon. You remember a runner and want to check whether you know her. All you know is that she arrived 37th on the finish line. Once the ranking with the names of participants is made available, how many bits do the organizers need to retrieve her name?

☐ 16 bits

☐ 11 bits

☒ 6 bits

☐ 5 bits



Answer

Correct:

We only need to represent the number 37, as it is sufficient for an algorithm to retrieve the name. According to the above coding, 37 can be represented with 6 bits: 100101. So six bits are sufficient to unambiguously designate the 37th runner at the finish line.

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The letter 'k' is a simple character in python.
What about the character '♖' (white chess rook)?
Its Unicode rank is 9814 (10011001010110 in binary form).
So we may estimate the complexity of '♖' as being about 14 (the length of 10011001010110).

Question: Can we assess the complexity of a character if we don't know its exact Unicode value?

Answer: Yes. Since there are a total of 1 114 112 different Unicode characters, we can be sure that the length of its binary representation must be less than 21 bits (you can check: `int('100010000000000000000', 2)` is 1114112, and `len('100010000000000000000')` is 21).

More generally:

When an object belongs to a set of size N ,
its complexity in that set cannot exceed
 $\log_2(N)$.

This is because $\log_2(N)$ bits are sufficient to represent any rank in the set of size N , ordered by whatever means.

Estimating the complexity of an object this way is useful whenever it belongs to a set that has no obvious order.
Consider for instance the cups in your grandmother's cupboard. They may be ordered from left to right and bottom to top (if stacked up), or by colour and size if they are different, and so on. We suppose that you wish to designate one of the cups using such an ordering method.



Unordered set

1/1 point (ungraded)

How many bits would you need to designate one specific cup in your grandmother's cupboard, if there is a total of 62 cups in it?

☐ 16 bits

☐ 11 bits

☒ 6 bits

☐ 5 bits



Answer

Correct:

We suppose that all cups are numbered using some unambiguous code. Since there are 62 cups, one is sure that the longest code cannot exceed 6 bits (as 62 can be represented as 111110). As long as the order (or, equivalently, the numbering method) and the cup are left unspecified, we cannot do any better (as compared with: "give me the largest cup").

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For now, what we know about measuring the complexity of an object is this:

- we must find a code to represent the object in the computer (if the object is stored in a list, the code can be the rank in the list),

- we express the code in binary form,
- we measure the length of this binary form.

But it's not sufficient. The notion of *complexity* is more demanding. One should find out the **shortest** possible binary code. So, *any code that can spare bits* is closer to the complexity of the entity.

Description Length within a set

0/1 point (ungraded)

Marie Martin is a French citizen you do not know. Her social security number is 2 36 07 24 322 986. Representing citizens by their social security number does not offer the maximally compact representation, though. Can you provide a good estimate of the complexity of Marie Martin, knowing that she is a living French citizen?

☒ 43 bits

☐ 35 bits

☐ 27 bits

☐ 22 bits

☐ 11 bits

☐ 8 bits

✖

Answer
Incorrect:
Marie Martin can be unambiguously determined by considering any numbering of all French Citizens (*e.g.* by increasing birth date) and by providing Marie Martin's rank in that list. Knowing that there are about $N = 67.2 \times 10^6$ people in France, we are sure that $\log_{2+}(N) = 27$ bits are sufficient to represent her. Note that this is much less than the logarithm of her social security number (if you answered $\log_{2+}(23624322986) = 35$ bits, don't worry, that's a typical overestimate!)

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Emoji complexity

1/1 point (ungraded)

Suppose the only emojis that are available on your phone app are just these:



How complex is the most complex emoji among them?

☒ 3 bits

☐ 8 bits

☐ 26 bits

☐ 127 bits

☐ 1264 bits



Answer

Correct:

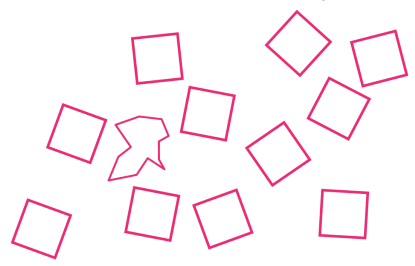
The complexity of objects within this set of emojis cannot exceed the log-size of the set. Here there are 7 emojis, so each of them can be retrieved with no more than 3 bits.

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 Answers are displayed within the problem

Consider these 12 objects.



The Ugly Duckling paradox

0/1 point (ungraded)

Which one is the simplest object in this set?

- ☒ One of the squares
- ☐ The odd shape



Answer

Incorrect:

The odd shape is the simplest object here, because it is unique. We need 4 bits to distinguish one square among the 11 squares. Strangely enough, we can say: "the simplest object is the most complex one." Of course, the odd shape would be complex if one wanted to describe it out of context. But in the present context, it is simple because it is just different.

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